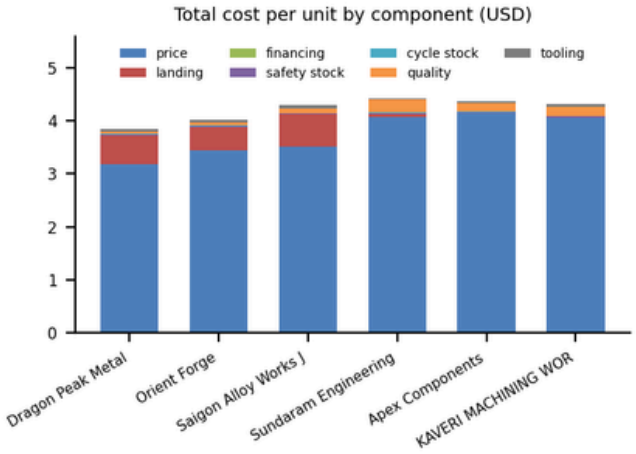
# Sourcing Copilot: supplier-quote comparison and disruption radar

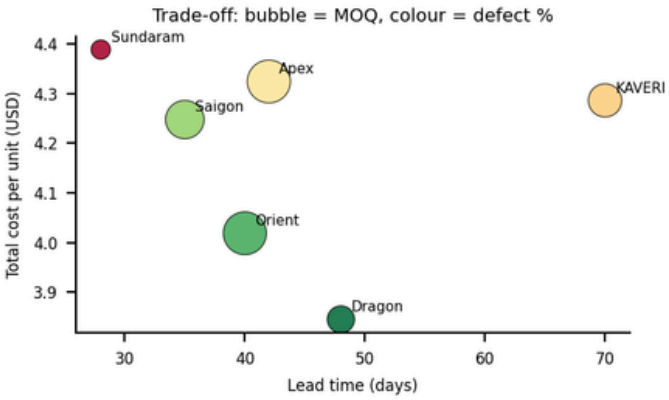
## What it does

**Job1,quote comparison.** Paste or upload free-text supplier quotes; the tool extracts price (with volume breaks), MOQ, lead time, payment terms, Incoterm, quality and tooling, shows every extracted value with its evidence, lets you correct it, and then compares suppliers on total cost per unit, lead time, volume flexibility and quality, with number-backed negotiation points. **Job 2, disruption radar.** News items are classified into nine event types with a direction (risk, easing, opportunity) and a 1-5 severity, then matched to a registry of mapped suppliers (the ten from the Project 2 dashboard plus three archetypes) to give a recency-weighted exposure score, the reason, the source link and a suggested action. A third tab lets live exposure reduce a supplier's decision score.

## Job 1 on the six sample quotes

Quoted prices span USD 3.18to 4.14, but total cost per unit spans 3.85 to 4.39 once landing cost (Incoterm and import duty), payment timing, MOQ carrying cost, quality and tooling are added. Dragon Peak Metal is cheapest at $3.85; KAVERI MACHINING WORKS is slowest at 70 days. The ranking under default weights (cost 50, lead time 20, MOQ 10, quality 20) puts Dragon Peak Metal first (score 88). This agrees with the Project 1 TCO model, where an import beats a local supplier on cost at base conditions, which the tool reproduces from the quotes alone.



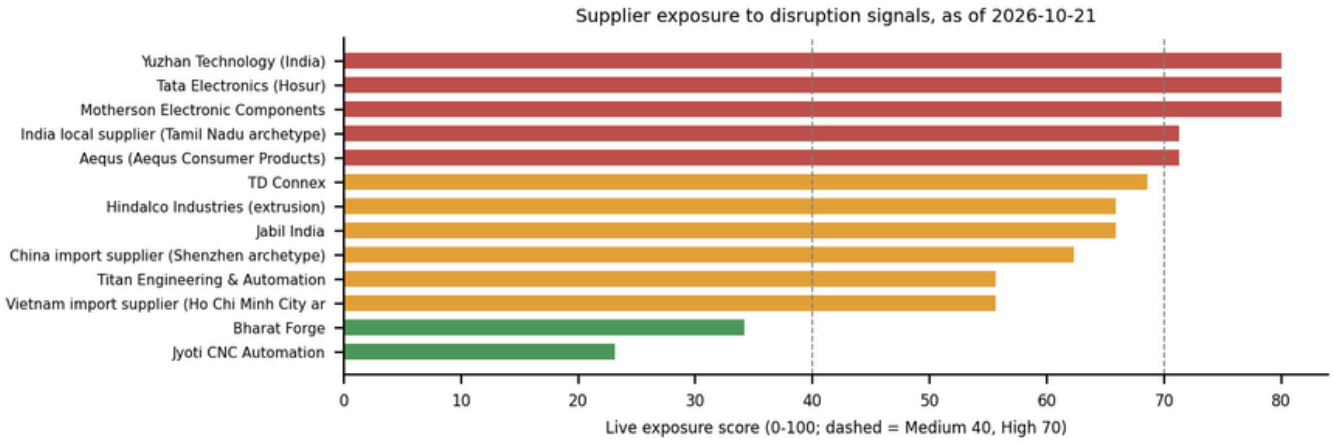




## Guards against silent errors

- **Grounding:** an LLM-extracted value is kept only if its evidence snippet literally appears in the quote; otherwise it is dropped and flagged.
- **Cross-check:** if rules and LLM disagree by more than 1% the field is flagged for review. Prices quoted per 1,000 units are normalised and flagged; prices far from peers are flagged.
- **No silent defaults:** missing fields use stated assumptions that are listed in the table and in the negotiation points. Every value is editable before scoring.

# Job 2 on the October 2026 news snapshot

Nineteen items(seventeen paraphrasedpublicreports and two invented noise items) were classified; 13 of 13 registry entries received non-zero exposure. 5 are rated High: Tata Electronics, Yuzhan Technology, Motherson Electronic Components, Aequs, India local supplier. The drivers are the Chennai-area freight and port signals (container rates up about 130% since February, Colombo feeder disruption affecting Chennai) and the ending of China's rare-earth control pause on 10 November, which touches suppliers with China-linked equipment dependence. Older events fade: the 2024 Hosur fire contributes almost nothing, the 2025 engineer recall less than 1%.




Risk adjustment did not change the quote ranking in this sample: the cost gap between the China and India quotes is wider than the difference in live exposure, and the ranking was unchanged for risk weights from 0 to 0.6. The tool reports that finding instead of forcing a story.

## Evaluation (author-labelled smoke test, not a benchmark)

| Engine | Set | Result |
|---|---|---|
| Quote fields (rules) | dev (3 quotes, tuned on) | 100% (33/33) |
| | test (3 quotes, not tuned) | 100% (33/33) |
| | hard (5 messy quotes) | 72% (38/53); 2 silent wrong numbers |
| News classification (rules) | dev (9 items) | type, direction, severity within 1, noise: all 100% |
| | test (10 items, not tuned) | type found 100%, exact primary 89%, direction 100%, severity 100%, noise 100% |
| LLM mode | - | implemented and tested with a fake client; **not run against the live API** (no key in the build environment) |

The hard set is the useful result: rules fail on amount-before-currency, one-line emails, '45 days credit' and summed lead-time phrases. That gap is the case for LLM mode with grounding, and the two silent wrong numbers show why every value needs evidence and a human-editable table.

## Limits and next steps

- Small, author-labelled sets; real quotes (PDF tables, scans) need a document parser in front of the extractor.
- Gateway ports, dependency flags, archetype risk and the exposure weights are analyst assumptions; they are editable but untested against real incidents.
- Live RSS fetch and LLM mode are implemented but were not run live here. Next: run `python evaluate.py llm` with a key, add PDF ingestion, and backtest exposure against past disruptions.

Run: pip install -r requirements.txt; streamlit run app.py (rules mode works offline); pytest -q (13 tests). See README.md.